

Integration Drill - Q21 to Q28 (coaching module sheet)

Answer with A-D or a number. Time: 8 minutes.

Q21. Integral of $2x \, dx$ is:

- (A) $x^2 + C$ (B) $2x^2 + C$ (C) $x + C$ (D) $2x + C$

Q22. $d/dx (\sin x) =$

- (A) $\cos x$ (B) $-\cos x$ (C) $\sin x$ (D) $\tan x$

Q23. Integral of $\sec^2 x \, dx$ is:

- (A) $\tan x$ (B) $\cot x$ (C) $\sec x$ (D) $-\tan x$

Q24. Value of integral of $3x^2 \, dx$ from 0 to 1 _____

Q25. If $f(x) = x^3$, $f'(2)$ equals _____

Q26. Integral of $1/x \, dx =$

- (A) $\ln|x| + C$ (B) $x^{-2/2} + C$ (C) $-1/x^2 + C$ (D) $e^x + C$

Q27. $d/dx (e^{2x}) =$

- (A) e^{2x} (B) $2e^{2x}$ (C) $2e^x$ (D) e^x

Q28. Area under $y = 2x$ from $x=0$ to $x=3$ _____